

1 The Mystery of the Missing Sandwich

A Story About Bacteria



It was Science Fair Week at Riverside School.

Lila and her best friend Jay were decorating their classroom.

Jay reached into his backpack.

“My sandwich!”

He had forgotten it there all weekend.

The bread smelled strange.

Tiny fuzzy patches had appeared.

Lila stepped back.

“Eww! What happened to it?”

Just then, their science teacher, Ms. Rivera, walked over with a smile.

“I think you’ve just discovered a tiny world that you can’t even see.”

Did You Know?

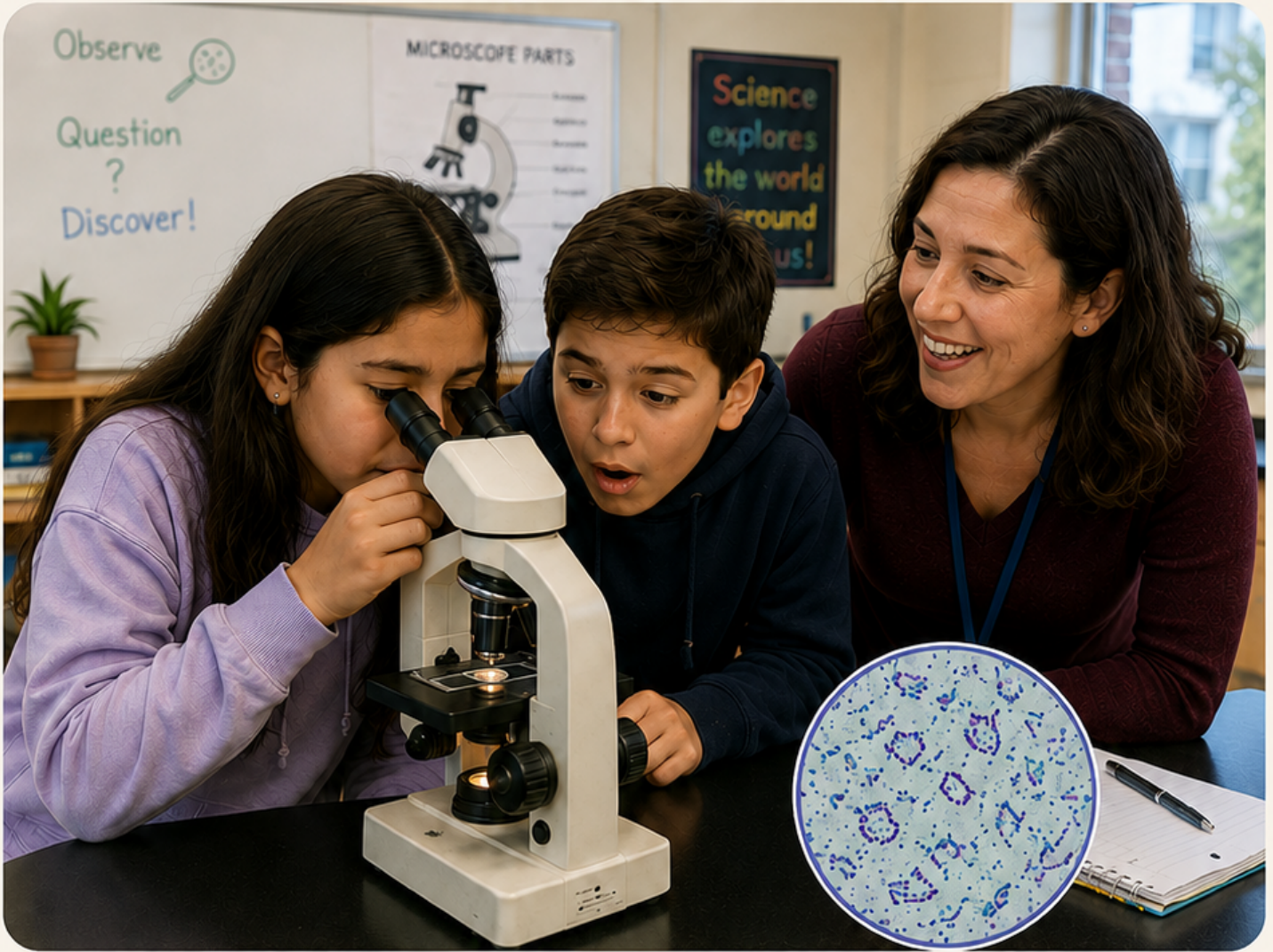
Bacteria are tiny living things. You need a microscope to see them!



2

The Invisible World

A Story About Bacteria



Ms. Rivera carefully placed a tiny crumb from the sandwich under a microscope. Lila looked through the eyepiece. “I see tiny little dots!” Jay took a turn. “There are thousands of them!” “Those are bacteria,” Ms. Rivera explained. “They are tiny living organisms. “They live almost everywhere—on plants, in soil, in water, and even on our skin.” “They’re so small that we need a microscope to see them.” Lila smiled. “It’s like an invisible city!”

Did You Know?

Bacteria are so tiny that millions of them can fit on the tip of your finger!



3 The Fastest Growing Town

A Story About Bacteria

BINARY FISSION

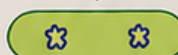
1. One bacterium



2. DNA copies



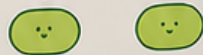
3. Cell stretches



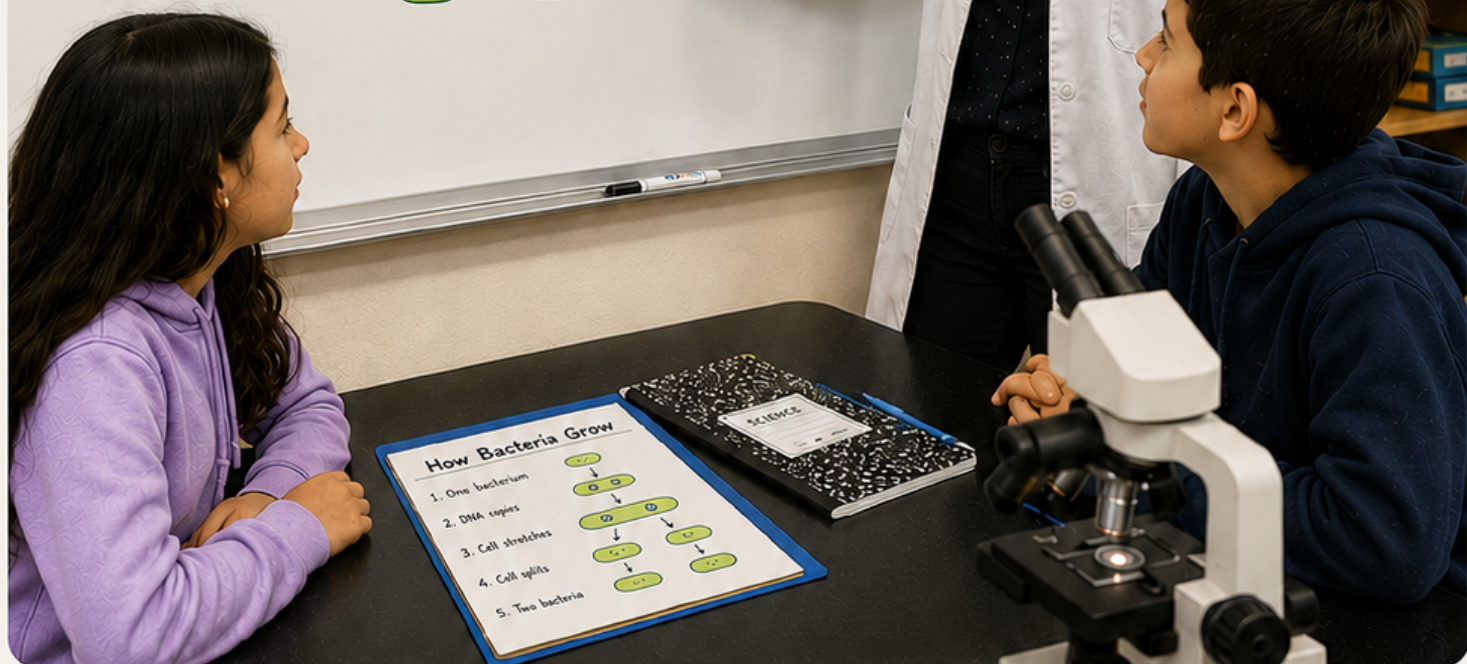
4. Cell splits



5. Two bacteria



This process
is called
binary fission.



Ms. Rivera drew one tiny circle on the whiteboard.

“Imagine this is one bacterium.”

“It copies everything inside itself.”

“Then it splits into two.”

“Each of those splits again.”

“And again.”

“This process is called binary fission.”

Jay’s eyes grew wide.

“So one tiny bacterium can become millions?”

“Exactly,” Ms. Rivera replied.

“That’s why food left out for too long
can quickly become covered with bacteria.”

Did You Know?

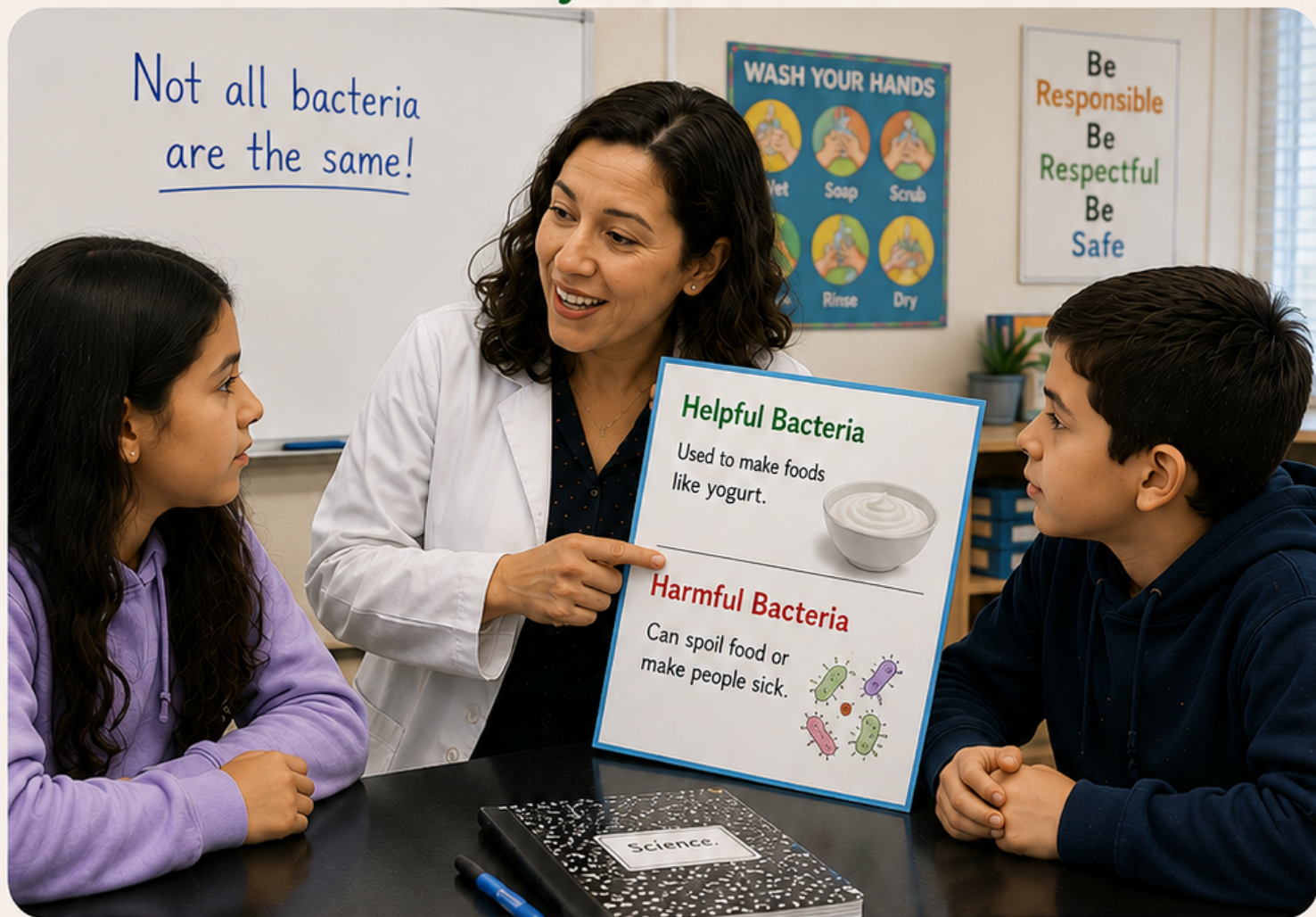
Bacteria reproduce very fast!

In the right conditions,
one bacterium can become
more than 1 million bacteria
in just 24 hours.



Good and Bad Neighbors

A Story About Bacteria



“Not all bacteria are the same,” Ms. Rivera explained.

“Many bacteria help us, like the ones used to make yogurt.”

“But some bacteria can spoil food or make people sick.”

“Bacteria can cause illnesses such as strep throat, staph infections, and some kinds of food poisoning caused by bacteria like *E. coli* and *Salmonella*.”

“Some bacteria can also cause dental cavities if we don’t brush our teeth.”

“So we wash our hands...”

“...keep food in the refrigerator...”

“...brush our teeth...”

“...and throw away spoiled food.”

Jay nodded.

“So bacteria aren’t the enemy.”

“We just need to stop the harmful ones.”

Did You Know?



Strep throat is caused by *Streptococcus* bacteria.



Staph infections are caused by *Staphylococcus* bacteria.



Food poisoning can be caused by *E. coli* and *Salmonella* bacteria.

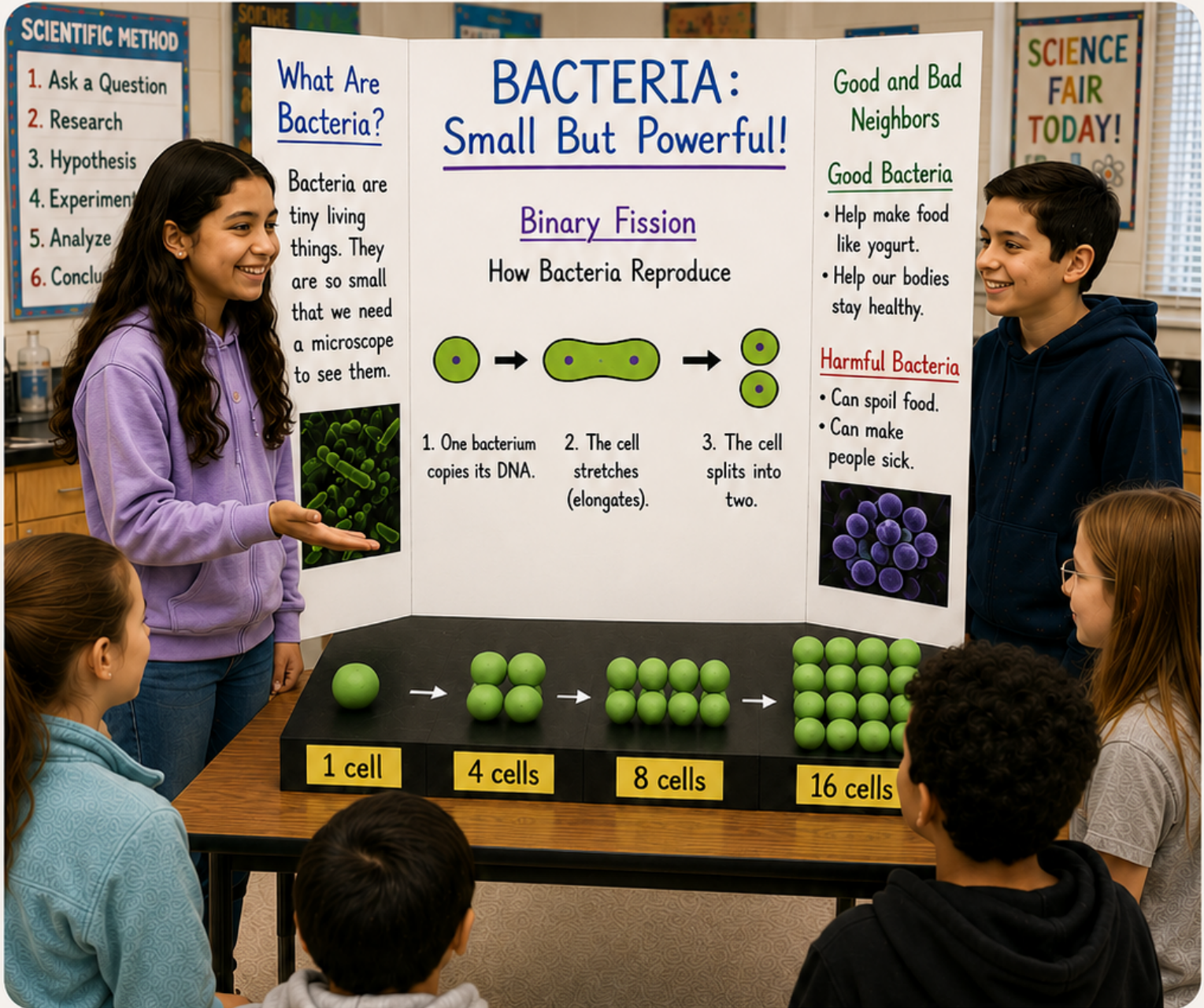


Dental cavities happen when bacteria in your mouth make acids that damage teeth.

Good habits help keep harmful bacteria away!

Tiny Scientists

A Story About Bacteria



For the Science Fair, Lila and Jay built a giant model of a bacterium. They also made a simple display showing binary fission.

One bacterium became two.

Two became four.

Four became eight.

Soon the chart was filled with tiny circles.

Their classmates gathered around.

“I never knew bacteria were living things,” one student said.

“They’re tiny...”

“...but they can have a big impact.”

Lila smiled.

“Sometimes the smallest discoveries are the biggest surprises.”

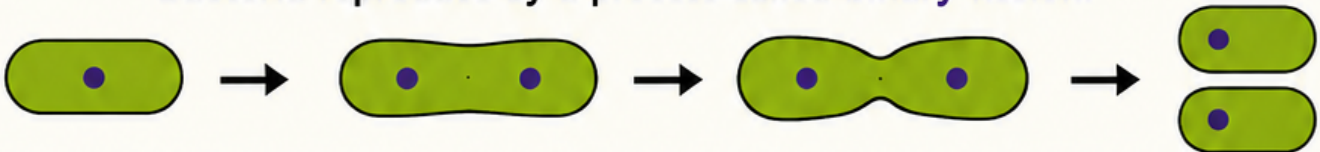
BACTERIA

- Bacteria are tiny single-celled living organisms.
- Millions of them can fit on the tip of your finger.
- Most bacteria are harmless, and many are helpful.
- Some bacteria can spoil food or make people sick.



HOW DO BACTERIA REPRODUCE?

Bacteria reproduce by a process called **binary fission**.



EXAMPLES OF BACTERIAL DISEASES



Strep throat
caused by
Streptococcus
bacteria.



Staph infections
caused by
Staphylococcus
bacteria.



Food poisoning
can be caused by
bacteria such as
E. coli and *Salmonella*.



Dental cavities
bacteria in your
mouth produce acids
that damage teeth.



HOW ARE BACTERIAL INFECTIONS TREATED?

- Antibiotics are medicines used to treat bacterial infections. They kill bacteria or stop them from growing.
- Antibiotics do not work against viruses, such as those that cause colds or the flu.

HOW CAN WE STAY HEALTHY?



Wash your
hands regularly.



Keep food in
the refrigerator.



Wash fruits and
vegetables before
eating.



Brush and floss
your teeth
every day.



Throw away
spoiled food.

★ REMEMBER

Bacteria are part of the invisible world all around us. Most are harmless, many are helpful, and learning how to stop harmful bacteria helps keep us healthy.

